Lemma. Let $\varphi : F \to F'$ be a

Let $p(x) \in F[x]$ be irredu

If β is a root of $p(x)$

then, there exists a unique

Proof. Since φ^* is also Iso

$$I : F[x]/(p(x)) \to F'[x$$

It maps $c + (p(x)) \mapsto \varphi(c) +$

Now, $\overline{\varphi} : F(\beta) \xrightarrow[\phi^{-1}]{} F[x]/(p(x)) \xrightarrow{I}$

Thm. A field of characteristi

Proof. Let $f(x) \in F[x]$ be a

$$\deg D_x f < \deg f.$$

Therefore, $(f, D_x f) = 1$, because

So it is separable.

Lemma. A field of characteristic

Proof Let F be a field of

Thm. Every finite separable e

Proof. If F is finite, t

Then, E^x is cyclic, thu

Now, assume that F is in

$$E = F(\alpha_1, \ldots, \alpha$$

Now, enough to show $F($

Let $\beta_1, \ldots, \beta_n$ be distinct

$\gamma_1, \ldots, \gamma_m$ be distinct